

IC 112: Calculus

Course Code	IC 112
Course Name	Calculus
L-T-P-C	1.5-0.5-0-2
Prerequisites	Consent of the faculty member
Intended for	B. Tech. 1st Year
Mutual Exclusion	(None)
Approval	50th BoA (Jul 2023); 42nd Senate (Jan 2024), in the IMBA (Integrated MBA) curriculum

Course Contents

- **Elementary calculus:** Real number system, Zeno's Paradox, Limit Continuity and Differentiability of single variables, Uniform continuity, Taylor series, Partial Derivatives. [6 Lectures]
- **Functions of Several Variables:** Limit, Continuity and differentiability of functions of two variables. Euler's Theorem, Tangent plane and Normal, Change of variables, Chain rule. Jacobians, Taylor's Theorem for Two Variables, Extrema of Functions of Two variables, Lagrange's method of undetermined multipliers. [8 Lectures]
- **Infinite Series:** Achilles' and Tortoise Problem, Sequences, Convergence of Infinite Series of Real Numbers, Comparison Test, Ratio Test, Root Test, Raabe's test, Logarithmic test, Demorgan's test, Sequence and series of functions: Uniform convergence and related tests. [7 Lectures]

Text Books

1. Thomas and Finney, **Calculus and Analytical Geometry**, 9th Edition, Addison and Wesley Publishing Company, 1996
2. W. Rudin, **Principles of Mathematical Analysis**.

References

1. E. Kreyszig, **Advanced Engineering Mathematics**, 10th Edition, Wiley.
2. J. E. Marsden, A. J. Tromba and A. Weinstein, **Basic Multivariable Calculus**, Springer, 1993.
3. Apostol, **Mathematical Analysis**, 2nd Edition.